

Student Management System Using MongoDB

A NoSQL mini project demonstrating core MongoDB operations on a collection of 30 student records, run in **mongosh**.

Project Details

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Overview

The project creates a Students database with a students collection. Each document stores a student's roll number, name, age, marks and city. The queries cover CRUD operations, comparison and logical operators, sorting, limiting and counting.

Document Structure

```
{
  roll: 101,
  name: "Shrishti Singh",
  age: 19,
  marks: 85,
  city: "Delhi"
}
```

Repository Contents

File	Description
README.pdf	Project overview and usage instructions
dataset.pdf	All 30 student records and the insertMany script
queries.pdf	All 18 MongoDB queries with explanations
No_sql_Project.pdf	Full project report with output screenshots

Operations Covered

Category	Operations
Database / Collection	use, createCollection
Create	insertMany
Read	find, countDocuments, limit, sort
Update	updateOne with \$set
Delete	deleteOne, deleteMany
Comparison operators	\$eq, \$gt, \$lt, \$gte, \$lte

Category	Operations
Logical operators	\$and, \$or

How to Run

1. Install MongoDB and mongosh.
2. Start the MongoDB server and open mongosh.
3. Run the commands from queries.pdf in order, using the data from dataset.pdf for the insert step.

Note on Results

Queries 11 and 12 delete records (roll 110, then rolls 111 and 113), and query 10 changes roll 105's marks to 95, so the collection holds 27 documents at the end.

Technologies Used

MongoDB, mongosh (MongoDB Shell)